Aim (8.1): WAP using try catch block. Users should enter two command line arguments. If only one argument is entered, then exception should be caught. In case of two Command line arguments, if fist is divided by second and if second command line argument is 0 then catch the appropriate exception.

```
public class Practical_8a{
   public static void main(String[] args) {
      try {
         if (args.length < 2) {
            throw new IllegalArgumentException("Error: Please enter exactly TWO command line arguments.");
         }

         int num1 = Integer.parseInt(args[0]);
         int num2 = Integer.parseInt(args[1]);

         if (num2 == 0) {
            throw new ArithmeticException("Error: Division by Zero is not allowed!");
         }

         int result = num1 / num2;
         System.out.println("First Argument  : " + num1);
         System.out.println("Second Argument : " + num2);
         System.out.println("Result (" + num1 + " / " + num2 + ") = " + result);

      } catch (NumberFormatException e) {
```

```
            System.out.println("Number Format Exception Caught: Arguments must be valid integers!");
        } catch (IllegalArgumentException e) {
            System.out.println("Argument Exception Caught: " + e.getMessage());
        } catch (ArithmeticException e) {
            System.out.println("Arithmetic Exception Caught: " + e.getMessage());
        } finally {
            System.out.println("\n[Finally Block]: Program execution complete.");
        }
    }
}
```

Output:

```
c:/Users/shn/Desktop/college_materials/sem_4/cjp/lab_practicals/Practical_8 $ java Practical_8a.java
Argument Exception Caught: Error: Please enter exactly TWO command line arguments.

[Finally Block]: Program execution complete.
c:/Users/shn/Desktop/college_materials/sem_4/cjp/lab_practicals/Practical_8 $ java Practical_8a.java 10 2
First Argument  : 10
Second Argument : 2
Result (10 / 2) = 5

[Finally Block]: Program execution complete.
c:/Users/shn/Desktop/college_materials/sem_4/cjp/lab_practicals/Practical_8 $ java Practical_8a.java 10 0
Arithmetic Exception Caught: Error: Division by Zero is not allowed!

[Finally Block]: Program execution complete.
c:/Users/shn/Desktop/college_materials/sem_4/cjp/lab_practicals/Practical_8 $ java Practical_8a.java abc 5
Number Format Exception Caught: Arguments must be valid integers!

[Finally Block]: Program execution complete.
c:/Users/shn/Desktop/college_materials/sem_4/cjp/lab_practicals/Practical_8 $
```

Aim (8.2): Define an exception called "NoMatchException" that is thrown when a string is not equal to "India". Write a program that uses this exception.

```
class NoMatchException extends Exception {
   public NoMatchException(String message) {
      super(message);
   }
}

public class Practical_8b {
   static void checkString(String input) throws NoMatchException {
      if (!input.equals("India")) {
         throw new NoMatchException("NoMatchException: \"" + input + "\" is not equal to
\"India\"");
      }
      System.out.println("Match found! The string is: " + input);
   }

   public static void main(String[] args) {
      String[] testInputs = {"India", "China", "India", "USA"};

      for (String s : testInputs) {
         try {
            checkString(s);
         } catch (NoMatchException e) {
            System.out.println(e.getMessage());
         }
      }
   }
}
```

Output:

```
c:/Users/shn/Desktop/college_materials/sem_4/cjp/lab_practicals/Practical_8 $ java Practical_8b.java
Match found! The string is: India
NoMatchException: "China" is not equal to "India"
Match found! The string is: India
NoMatchException: "USA" is not equal to "India"
c:/Users/shn/Desktop/college_materials/sem_4/cjp/lab_practicals/Practical_8 $
```